

Appendix A – BMS data CAN bus messages

All data will be send in "Big Endian" format.

Signed data is formatted as two's complement.

CAN bus start address (N) and bitrate can be programmed with the 123\SmartBMS Extended Module PC Software

Address:	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
N+0	Total voltage		Current IN		Current OUT		Current Battery	
N+1	Energy stored				Battery capacity		SOC	-
N+2	Energy today collected				Energy today consumed			
N+3	Total energy collected				Total energy consumed			
N+4	Cell voltage MIN		Cell voltage MAX		Cell voltage Bypass		-	-
N+5	Cell voltage Lowest		Low Nr	Cell voltage Highest		High Nr	Sbyte 1	Sbyte2
N+6	Tmp,Lowest	Low Nr	Tmp,Highest	High Nr	Min charg temp	Min dis temp	Max temp	-
N+7	Current cell voltage		Cur.Temp	Cur. Nr	Cell cnt	-	Isolation resistance	

Message	Step	Size	Signed	Example 1	Example 2
Total Voltage	0.1 Volt/bit	16 bit	-	0x15FF = 563.1V	
Current IN	0.1 Amp/bit	16 bit	Signed	0x0230 = 56.0 Amp	0xFF6E = -14.6 Amp
Current OUT	0.1 Amp/bit	16 bit	Signed	0x0230 = 56.0 Amp	0xFF6E = -14.6 Amp
Current Battery	0.1 Amp/bit	16 bit	Signed	0x0230 = 56.0 Amp	0xFF6E = -14.6 Amp
Energy stored	Wh/bit	32 bit	-	0x00A3 = 163 kWh	
Batery capacity	0.1 kWh/bit	16 bit	-	0x00A0 = 16.0 kWh	
SOC (state of charge)	1%/bit	8 bit	-	0x32 = 50%	
Energy today collected	Wh/bit	32 bit	-	0x64 = 100 Wh	
Energy today consumed	Wh/bit	32 bit	-	0x64 = 100 Wh	
Total energy collected	kWh/bit	32 bit	-	0x00A3 = 163 kWh	
Total energy consumed	kWh/bit	32 bit	-	0x00A3 = 163 kWh	
Cell voltage MIN	1mV/bit	16 bit	-	0x09C4 = 2.500V	OUT Board settings
Cell voltage MAX	1mV/bit	16 bit	-	0x0E74 = 3.700V	OUT Board settings
Cell voltage Bypass	1mV/bit	16 bit	-	0x0DAC = 3.500V	OUT Board settings
Cell Voltage lowest	1mV/bit	16 bit	-	0x0B86 = 2,950V	
Low nr.	Nr/bit	8 bit	-	0x64 = cell nr 100	
Cell Voltage highest	1mV/bit	16 bit	-	0x0E10 = 3,600V	
High nr.	Nr/bit	8 bit	-	0x32 = cell nr 50	
Cell temp lowest	1°C/bit	8 bit	Signed	0x14 = 20 °C	0xFA = -6 °C
Low nr.	Nr/bit	8 bit	-	0x16 = cell nr 22	
Cell temp highest	1°C/bit	8 bit	Signed	0x14 = 20 °C	0xFA = -6 °C
High nr.	Nr/bit	8 bit	-	0x16 = cell nr 22	
Minimum charging	1°C/bit	8 bit	Signed	0x14 = 20 °C	

temperature								
Minimum discharge temp	1°C/bit		8 bit	Signed	0x15 = 21 °C			
Maximum temperature	1°C/bit		8 bit	Signed	0x16 = 20 °C			
Current cell voltage	1mV/bit		16 bit	-	0x0B86 = 2,950V			
Current cell temperature	1°C/bit		8 bit	Signed	0x14 = 20 °C	0xFA = -6 °C		
Current nr.	Nr/bit		8 bit	-	0x18 = cell nr 24			
Cell count	Nr/bit		8 bit	-	0xFF = 255 cells			
Isolation resistance	kΩ/bit		16 bit	-	0x258 = 600kΩ			

Status byte	Bit	Parameter
Status byte 1	MSB	Allow to charge
	6	Allow to discharge
	5	Cell communication error
	4	BMS communication error
	3	Exceed minimum voltage
	2	Exceed maximum voltage
	1	Exceed minimum temperature
	LSB	Exceed maximum temperature
Status byte 2	MSB	-
	6	J1772 = connected
	5	J1772/Elcon/TC = charging
	4	-
	3	-
	2	-
	1	-
	LSB	-